

FINAL PROJECT PRESENTATION

AUTONOMOUS ROBOT COMPONENTS

CONTROLLER [Arduino UNO]

- reliable
- have 14 digital I/O pins, able to connect to many sensors, motor and servo

[Motor Driver]

- have two channels that are perfect for two driving motor
- able to control the speed of each motor for stability

DRIVING [TT Gear DC Motor]

- lightweight
- enough torque to run smoothly
- need motor driver and battery supply to be able to run

COLLECTOR [SG90 micro servo]

- lightweight with only 9 g
- with enough torque of two servo that be able to push a 70mm cube

GYRO SENSOR [MPU6050]

- it's a combination of gyroscope and the accelerometer
- when uses with motor driver, we can control the speed of each motor to stable the driving

ULTRASONIC [HC-SR04]

- have a range of 20mm to 4000mm
 - have resolution of about 3mm
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MANUAL ROBOT COMPONENTS

CONTROLLER [mBot2 + CyberPi]

- it support WiFi and Bluetooth 5.0

- able to control it with makeblock controller

DRIVING [Encoder Motor]

- it can rotate 360 degrees
- torque of 0.8 kg cm
- be able to control the power and speed

GRIPPER [MS-1.5A Servo]

- rotates 180 degrees
- torque of 1.8 kg cm that be able to grab different size balls

LIFTER [Digital Servo]

- has a really high torque, over 20 kg cm
- be able to handle the arm, two small servo, and the ball all together

STRIKER [TT Gear DC Motor]

- has a speed of 200 RPM that generate torque of about 0.8 kg cm
- this motor generate speed enough to strike a ball to a prefreable range.